

ENGINEERING PROJECT

DESIGN AND SIMULATION OF A DIGITAL DOOR ACCESS CONTROL SYSTEM

A dependable credential-to-lock decision

AN INITIATIVE PROJECT BY

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 **Guide:** ChatGPT

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ACCESS CONTROL PROTECTS PEOPLE, ASSETS, AND INFORMATION

A digital door verifies an authorized user before releasing the lock.
Security becomes a repeatable decision—not a judgment made at the door.



User credential

Input is captured.



Controller verification

Stored value is
checked.



Lock status

Unlock or alarm
follows.

02 • THE GAP

TRADITIONAL LOCKS LEAVE A GAP FOR CONTROLLED, TRACEABLE ACCESS

The design need is simple: authenticate users, control the door electronically, and signal denied access.

DEFINITION

Authorized match

An access control system permits entry only when the presented credential matches an authorized value.

LIMITATION

Physical keys

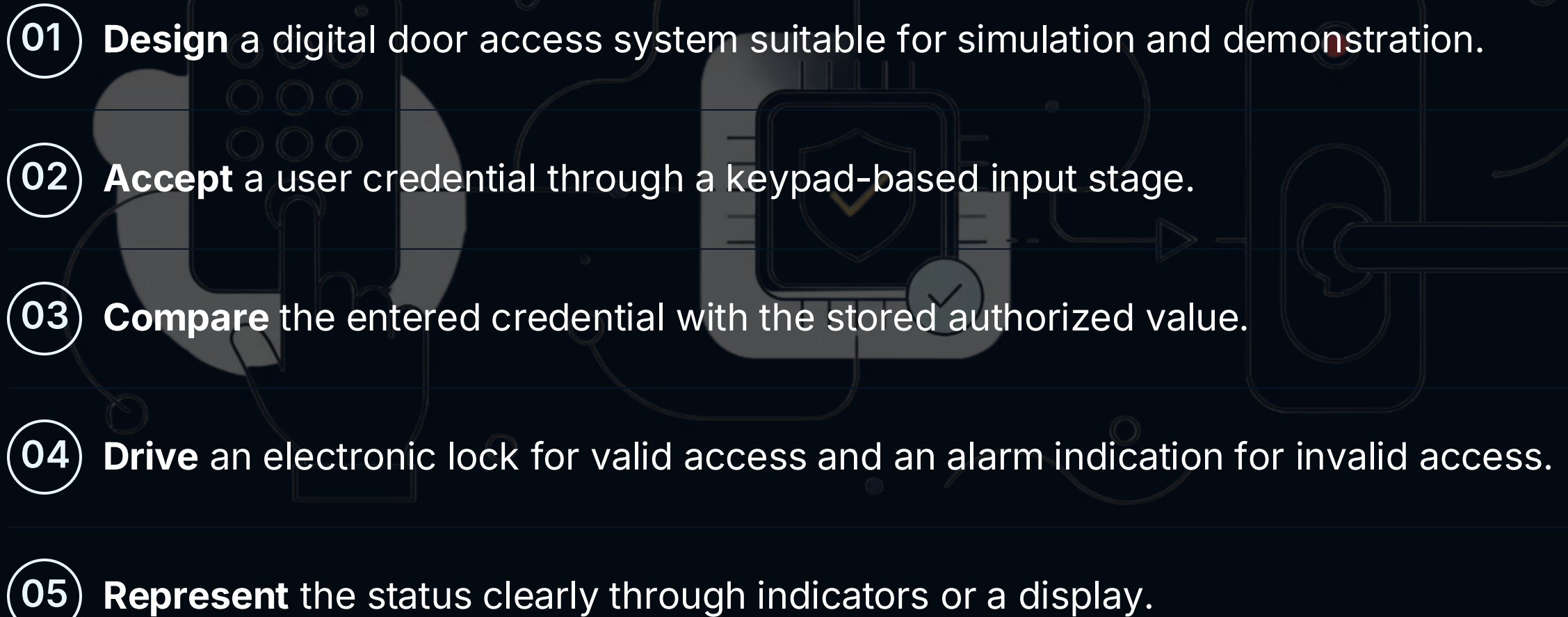
Keys can be lost, copied, or shared; manual checking is slow and inconsistent.

NEED

Digital response

One compact system should authenticate, actuate, and alert.

THE PROJECT TURNS AUTHENTICATION INTO A DEPENDABLE DECISION

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- The background features a stylized illustration of a digital door access system. On the left, a hand is shown entering a PIN into a keypad. A line connects this to a central unit with a shield icon and a checkmark, representing the authentication process. Another line leads from the central unit to a door handle on the right, indicating the outcome of the authentication. The entire scene is set against a dark blue background with faint geometric shapes and lines.
- 01 **Design** a digital door access system suitable for simulation and demonstration.
 - 02 **Accept** a user credential through a keypad-based input stage.
 - 03 **Compare** the entered credential with the stored authorized value.
 - 04 **Drive** an electronic lock for valid access and an alarm indication for invalid access.
 - 05 **Represent** the status clearly through indicators or a display.

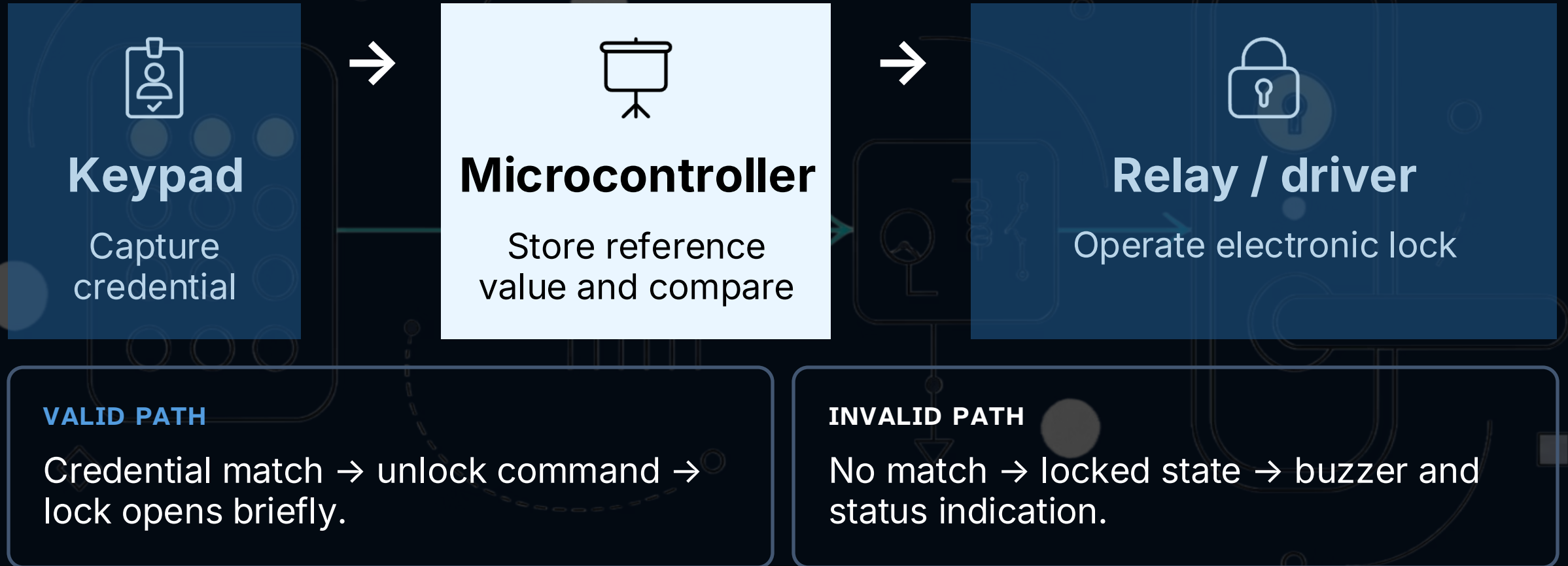
04 · EXISTING SYSTEMS

EXISTING SYSTEMS TRADE CONVENIENCE FOR CONTROL

The project gap is a compact design that combines authentication, actuation, feedback, and alert logic.

System	Strength	Limitation	What is missing
Mechanical key	Simple	Key can be lost or copied	Controlled, traceable entry
Manual checking	Human judgment	Slow and inconsistent	Repeatable decision logic
Basic electronic lock	Remote or digital control	Limited feedback and logic	Clear valid/invalid response

THE PROPOSED SYSTEM SEPARATES **INPUT**, LOGIC, AND ACTUATION



Simulation view: observe the signal path from credential entry to unlock or alarm.

FIVE MODULES MAKE THE DESIGN **EASY TO TEST**

1

Credential input

Captures the entry.



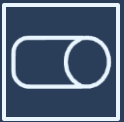
Controller

Stores, processes, and decides.



Display & indicators

Shows prompts and status.



Lock driver

Isolates and switches.

Alert & reset

Signals denial and restores safe start.

07 · WORKING PRINCIPLE

DESIGN THE CIRCUIT, THEN VERIFY THE ACCESS LOGIC

Build the signal path first. Test both decision branches before the next attempt.

Design steps

Define flow

Connect circuit

Simulate tests

Design steps: define the credential flow → connect input and controller → add display and indicators → add driver and lock → simulate test cases.

VALID

Read credential → compare with stored value → if valid, energize lock for a timed interval.

INVALID

Read credential → compare with stored value → otherwise keep locked and sound the buzzer.

Circuit components: keypad, microcontroller, LCD or status display, relay/transistor driver, electronic lock, buzzer, LEDs, resistors, regulated supply, and reset switch.

Logic: read credential → compare with stored value → if valid, energize lock for a timed interval → otherwise keep locked and sound the buzzer → reset for the next attempt.

08 • CONCLUSION

A DIGITAL ACCESS SYSTEM MAKES THE SECURITY DECISION CONSISTENT

Authenticate the user. Decide with logic. Actuate safely.



Authenticate

Replace manual checks with repeatable credential verification.



Decide

Separate input, control logic, and lock actuation for easy simulation, testing, and improvement.



Actuate

Valid credentials open the door; invalid credentials keep it locked and trigger an alert.

Key takeaway: a compact digital controller can deliver controlled, understandable, and scalable door access.